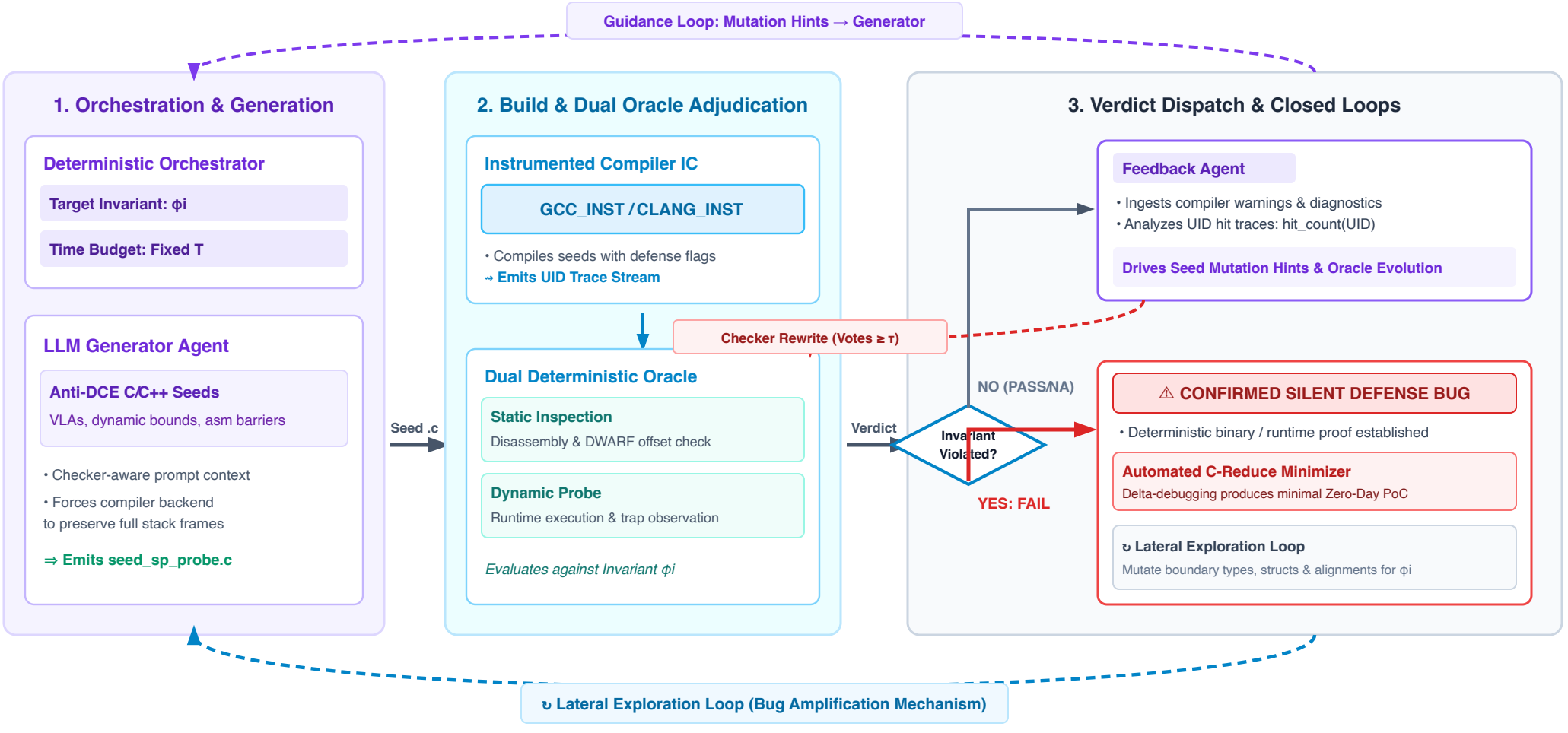

Guidance Loop: Mutation Hints → Generator
1. Orchestration & Generation
Deterministic Orchestrator
Target Invariant: ϕi
Time Budget: Fixed T
LLM Generator Agent
Anti-DCE C/C++ Seeds
VLAs, dynamic bounds, asm barriers
• Checker-aware prompt context
• Forces compiler backend to preserve full stack frames
⇒ Emits seed_sp_probe.c
Seed .c
2. Build & Dual Oracle Adjudication
Instrumented Compiler IC
GCC_INST / CLANG_INST
• Compiles seeds with defense flags
↠ Emits UID Trace Stream
Checker Rewrite (Votes ≥ τ)
Dual Deterministic Oracle
Static Inspection
Disassembly & DWARF offset check
Dynamic Probe
Runtime execution & trap observation
Evaluates against Invariant ϕi
Verdict
Invariant Violated?
NO (PASS/NA)
YES: FAIL
3. Verdict Dispatch & Closed Loops
Feedback Agent
• Ingests compiler warnings & diagnostics
• Analyzes UID hit traces: hit_count(UID)
Drives Seed Mutation Hints & Oracle Evolution
⚠ CONFIRMED SILENT DEFENSE BUG
• Deterministic binary / runtime proof established
Automated C-Reduce Minimizer
Delta-debugging produces minimal Zero-Day PoC
↻ Lateral Exploration Loop
Mutate boundary types, structs & alignments for ϕi
↻ Lateral Exploration Loop (Bug Amplification Mechanism)